

Question number	Answer	Additional guidance	Mark
7(a)(i)	A description that includes four of the following points: • impulse (arriving) at the pre-synaptic knob opens (voltage-gated) calcium ion channels (1) • calcium ions diffuse in and cause vesicles containing neurotransmitter to fuse with the presynaptic membrane (1) • neurotransmitter is released into the synaptic cleft and binds to receptors on postsynaptic membrane (1) • opening (ligand-gated) sodium channels (1) • depolarising the membrane and initiating an action potential (1)		(4)

Question number	Answer	Additional guidance	Mark
7(a)(ii)	A description that includes three of the following points: • (in Parkinson's disease) dopamine producing cells in the motor cortex die (1) • L-DOPA crosses the blood brain barrier (1) • in the brain L-DOPA is converted to dopamine (1) • restoring control of muscle movement (1)	ACCEPT in basal ganglia / in midbrain ACCEPT replacing the dopamine lost in the motor cortex	(3)

Question number	Answer	Additional guidance	Mark
7(b)(i)	An explanation that includes two of the following points: • (transcription factors) bind to {DNA / genes} involved in the synthesis of dopamine(1) • allowing mRNA synthesis (1) • that is translated to produce the enzymes that synthesise dopamine (1)	ACCEPT gene/promotor region of DNA for dopamine ACCEPT RNA polymerase binding / forming a transcription initiation complex	(2)

Question number	Answer	Additional guidance	Mark
7(b)(ii)	An answer that includes the following points: • both mean number and length of dendrites decrease in LRRK2 and Parkinson's Disease neurones compared with the control neurones (1) • the error bars do not overlap between people without Parkinson's disease and those with Parkinson's disease (1) • the error bars overlap between Parkinson's disease and LRRK2 neurones (1) • the number of dendrites are the same for both Parkinson's groups but the length is longer in the LRRK2 group	ALLOW the difference is statistically significant ALLOW the differences are not statistically different between different causes of Parkinson's	(3)

